The Price of Growth

Design Report: Visualization Choices and Narrative Structure

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1. Topic and motivation

I wanted to explore which US sectors actually create value for investors and which ones just look like they do. A company or sector can be huge and profitable on paper, but if its return on capital does not beat its cost of capital, it is really destroying value. I found Aswath Damodaran's January 2026 dataset on NYU Stern, which covers 94 US industries with all the financial metrics I needed: market cap, WACC, ROIC, EVA, growth rates, and capital allocation data. The dataset gave me everything in one place, already aggregated at the industry level, which meant I could focus on the story instead of spending weeks cleaning raw company filings.

2. Narrative structure

I used a Freytag pyramid to structure the narrative. Charts 1 and 2 set the stage: how big is each sector, and what does capital cost? Chart 3 is the climax because it directly answers the central question of who creates value and who destroys it. Chart 4 checks whether the stock market agrees with those fundamentals. Chart 5 wraps things up by showing what winning industries do with their profits, connecting the abstract concept of value creation back to concrete capital allocation decisions. Charts 1, 3, and 4 show industries individually (between 85 and 94 data points). Chart 2 aggregates into 11 GICS sectors because 94 individual WACCs would be unreadable. Chart 5 focuses on 12 hand-picked industries from the four quadrants of Chart 3.

3. Chart-by-chart design choices

Chart 1: Sector landscape (treemap)

I needed to show two things at once: how large each industry is (market cap) and how profitable it is (net margin). Tile area encodes market cap, and fill color uses a diverging scale from vermillion through warm gold and pale yellow to sky blue and deep blue, following the Okabe-Ito palette. Negative margins sit in the warm end and high margins in the cool end, which avoids the typical red to green diverging scale that is invisible to people with deuteranopia or protanopia. The treemap works as exposition because it orients the reader without making claims about value creation yet.

Chart 2: Cost of capital (stacked bar)

Each bar breaks down the WACC into its debt piece (Kd times D/V) and equity piece (Ke times E/V), weighted by market cap across all industries in the sector, sorted from lowest to highest. The two segments use sky blue (#56B4E9) for debt and deep blue (#0072B2) for equity. I chose two shades from the same blue family so they read as parts of a whole while remaining distinguishable under all three types of color vision deficiency. This is the rising action: if Technology has a 9.5% hurdle, the natural question becomes whether it earns enough to clear it.

Chart 3: Value creation (bubble chart)

This is the core of the story. The x-axis shows the ROIC minus WACC spread, the y-axis shows expected EBIT growth (g = ROC times reinvestment rate), and bubble size encodes absolute EVA. Blue bubbles (#0072B2) mean the industry creates value, orange (#E69F00) means it destroys it. I chose this pair because it is the highest contrast combination in the Okabe-Ito palette and works for deuteranopia, protanopia, and tritanopia. Four quadrants emerge: compounders (top-right), cash cows (bottom-right), growth traps (top-left), and value traps (bottom-left).

To make the chart readable for non-experts, I added two-level quadrant labels: a title (for example “COMPOUNDERS”) plus a plain-language sub-description (“Growing fast and creating value”). Someone who has never heard of ROIC can still understand what each zone means. The axis titles include directional helpers (“higher = more value created”) in subtle, low opacity text so they do not clutter the chart for expert readers. I considered using implied valuation multiples (EV/IC = (ROIC minus g) / (WACC minus g)) but rejected the approach because it assumes constant perpetual growth, which does not hold across 94 industries.

Chart 4: Market’s verdict (scatter)

I plotted EV/IC against the ROIC minus WACC spread to see if the market prices value creation correctly. The OLS regression gives r = 0.55; the interesting part is not the trend line but the outliers: Software (Internet) has a negative spread but trades at 9x EV/IC (the market bets on improvement), while Tobacco has a +56 pp spread at only 11x (the market sees no growth). For the 11 sector colors I combined Okabe-Ito with the Tol qualitative scheme, giving 11 hues distinguishable under deuteranopia and protanopia.

I added zone annotations (“Market pays a premium for value creation” in the top-right, “Destroying value, market discounts it” in the bottom-left) and axis helpers (“right = earns above cost of capital,” “higher = market pays more per dollar”) to orient non-expert readers. I chose EV/IC over EV/EBITDA or EV/NOPAT after testing all three: EV/IC had the strongest correlation (r = 0.55, R^2 = 0.35), while EV/NOPAT was negatively correlated (r = -0.33) and EV/EBITDA was essentially flat (r = -0.13). One caveat: EV/IC favors asset-light industries where R&D is expensed rather than capitalized.

Chart 5: Capital allocation (stacked bar)

I picked 12 industries (4 compounders, 4 cash cows, 4 destroyers) and decomposed net income into dividends, buybacks, and retained earnings. The four color segments are warm gold (#E69F00) for dividends, sky blue (#56B4E9) for buybacks, deep blue (#0072B2) for retained earnings, and vermillion (#D55E00) for excess payout. Blue consistently means “positive” and warm tones mean “negative” across all five charts. I added group labels on the right margin (Compounders, Cash Cows, Destroyers) so readers see which group each industry belongs to without referencing Chart 3. The axis says “how each industry splits its profits” instead of just “% of Net

Income." Each bar has a spread annotation (blue for positive, orange for negative) tying the allocation pattern back to the value creation metric from Chart 3.

4. Color palette and accessibility

The entire project uses the Okabe-Ito palette, an academic standard for colorblind safe visualization. Its eight colors are chosen so that any pair remains distinguishable under deuteranopia, protanopia, and tritanopia. About 8% of men and 0.5% of women have some form of color blindness, so this was not optional. I kept a consistent semantic mapping across all charts: blue (#0072B2) means positive (value creation, reinvestment, high margin) and warm tones (#E69F00, #D55E00) mean negative (value destruction, excess payout, low margin). Chart 1 uses a full diverging gradient. Chart 2 uses two blue shades for a neutral cost decomposition. Chart 3 uses the highest contrast blue/orange pair. Chart 4 extends to 11 colors via the Tol scheme. Chart 5 uses four Okabe-Ito colors for the allocation buckets.

Beyond color, I made sure it is never the only channel carrying information. Chart 3 uses bubble size and quadrant position. Chart 4 has a regression line and zone annotations. Chart 5 has group labels and spread annotations. Even if someone could not perceive the color difference at all, position, size, and text cues still communicate the core message.

5. Accessibility for non-experts

ROIC, WACC, EVA, and EV/IC mean nothing to someone outside of finance. I addressed this at three levels. First, Chart 3's quadrant labels use two lines: a title ("COMPOUNDERS") plus a plain language explanation ("Growing fast and creating value"). Second, axis titles across Charts 3, 4, and 5 carry directional helpers ("higher = more value created," "how each industry splits its profits"). Third, Chart 4's zone annotations explain what the scatter regions mean in plain terms. All annotations use low opacity and small fonts, so they stay in the background for expert readers but remain available for anyone who needs them.

6. Tools and implementation

I used Python 3.10, Pandas for data loading and aggregation, NumPy for the OLS regression, and Plotly graph_objects for all five charts (go.Treemap, go.Bar, go.Scatter). Development happened in a Jupyter notebook (notebooks/01_data_story.ipynb) and the final interactive version runs as a Streamlit app (app.py). Both files share the same data loading and chart logic. All data comes from seven CSVs published by Damodaran at NYU Stern (January 2026 update).